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compact1, compact2, compact3

java.util

Class Dictionary<K,V>

java.lang.Object
    java.util.Dictionary<K,V>

Direct Known Subclasses:

Hashtable

```
public abstract class Dictionary<K,V>
extends Object
```

The `Dictionary` class is the abstract parent of any class, such as `Hashtable`, which maps keys to values. Every key and every value is an object. In any one `Dictionary` object, every key is associated with at most one value. Given a `Dictionary` and a key, the associated element can be looked up. Any non-`null` object can be used as a key and as a value.

As a rule, the `equals` method should be used by implementations of this class to decide if two keys are the same.

NOTE: This class is obsolete. New implementations should implement the Map interface, rather than extending this class.

Since:

`JDK1.0`

See Also:

`Map`, `Object.equals(java.lang.Object)`, `Object.hashCode()`, `Hashtable`

Constructor Summary

Constructors

Constructor	Description

Dictionary()	Sole constructor.

Method Summary

Abstract Methods

Modifier and Type	Method	Description
abstract **Enumeration**<**V**>	**elements**()	Returns an enumeration of the values in this dictionary.
abstract **V**	**get**(**Object** key)	Returns the value to which the key is mapped in this dictionary.
abstract boolean	**isEmpty**()	Tests if this dictionary maps no keys to value.
abstract **Enumeration**<**K**>	**keys**()	Returns an enumeration of the keys in this dictionary.
abstract **V**	**put**(**K** key, **V** value)	Maps the specified key to the specified value in this dictionary.
abstract **V**	**remove**(**Object** key)	Removes the key (and its corresponding value) from this dictionary.
abstract int	**size**()	Returns the number of entries (distinct keys) in this dictionary.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

Dictionary

```
public Dictionary()
```

Sole constructor. (For invocation by subclass constructors, typically implicit.)

Method Detail

size

```
public abstract int size()
```

Returns the number of entries (distinct keys) in this dictionary.

Returns:

the number of keys in this dictionary.

isEmpty

```
public abstract boolean isEmpty()
```

Tests if this dictionary maps no keys to value. The general contract for the `isEmpty` method is that the result is true if and only if this dictionary contains no entries.

Returns:

true if this dictionary maps no keys to values; false otherwise.

keys

```
public abstract Enumeration<K> keys()
```

Returns an enumeration of the keys in this dictionary. The general contract for the keys method is that an `Enumeration` object is returned that will generate all the keys for which this dictionary contains entries.

Returns:

an enumeration of the keys in this dictionary.

See Also:

`elements()`, `Enumeration`

elements

```
public abstract Enumeration<V> elements()
```

Returns an enumeration of the values in this dictionary. The general contract for the `elements` method is that an `Enumeration` is returned that will generate all the elements contained in entries in this dictionary.

Returns:

an enumeration of the values in this dictionary.

See Also:

`keys()`, `Enumeration`

get

```
public abstract V get(Object key)
```

Returns the value to which the key is mapped in this dictionary. The general contract for the `isEmpty` method is that if this dictionary contains an entry for the specified key, the associated value is returned; otherwise, `null` is returned.

Parameters:

`key - a key in this dictionary. null if the key is not mapped to any value in this dictionary.`

Returns:

the value to which the key is mapped in this dictionary;

Throws:

`NullPointerException - if the key is null.`

See Also:

`put(java.lang.Object, java.lang.Object)`

put

```
public abstract V put(K key,
                      V value)
```

Maps the specified `key` to the specified `value` in this dictionary. Neither the key nor the value can be `null`.

If this dictionary already contains an entry for the specified `key`, the value already in this dictionary for that `key` is returned, after modifying the entry to contain the new element.

If this dictionary does not already have an entry for the specified `key`, an entry is created for the specified `key` and `value`, and `null` is returned.

The `value` can be retrieved by calling the `get` method with a `key` that is equal to the original `key`.

Parameters:

`key - the hashtable key.`

`value - the value.`

Returns:

`the previous value to which the key was mapped in this dictionary, or null if the key did not have a previous mapping.`

Throws:

`NullPointerException - if the key or value is null.`

See Also:

`Object.equals(java.lang.Object)`, `get(java.lang.Object)`

remove

```
public abstract V remove(Object key)
```

Removes the `key` (and its corresponding `value`) from this dictionary. This method does nothing if the `key` is not in this dictionary.

Parameters:

`key - the key that needs to be removed.`

Returns:

`the value to which the key had been mapped in this dictionary, or null if the key did not have a mapping.`

Throws:

`NullPointerException` - if key is null.

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